

User Guide: Interactive Quadratic Equation Solver

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Contents

1	Introduction	1
2	System Requirements	1
3	Setup Instructions	1
4	Running the Program	2
4.1	Mode 1: Direct Console Input	2
4.1.1	Example 1: Valid Input	2
4.1.2	Example 2: Invalid Input	2
4.2	Mode 2: File Input	3
4.2.1	Example Run with File Input	3
5	Troubleshooting	3

1 Introduction

Welcome to the Interactive Quadratic Equation Solver! This program is a tool for finding the roots of equations in the form $ax^2 + bx + c = 0$. You can provide the coefficients a , b , and c either by typing them directly or by placing them in a file. The program will validate your input and solve the equation for you.

2 System Requirements

- **A C Compiler:** You need a compiler like **GCC** to build the program from the source code.
- **A Plain Text Editor:** To write the code and create the input file (e.g., Notepad, VS Code).

3 Setup Instructions

Step 1: Create the Program File

Copy the entire C source code and save it in a new folder as `solver.c`.

Step 2: Compile the Program

Open a terminal or command prompt, navigate to your folder, and run the following command:

```
gcc solver.c -o solver_app -lm
```

Note: The `-lm` flag is very important. It links the math library, which is required for the program to work correctly.

4 Running the Program

The program has two modes of operation, which you select when you start it.

4.1 Mode 1: Direct Console Input

This is the most straightforward way to use the solver.

Step 1: Run the Executable

In your terminal, type: `./solver_app`

Step 2: Choose the Input Method

The program will prompt you to choose an input method. Type 1 and press Enter.

Step 3: Enter Coefficients

The program will ask you for the values of a , b , and c . Type each one and press Enter.

4.1.1 Example 1: Valid Input

```
$ ./solver_app
Select Input Method:
  1: Direct console input
  2: File input (from coeffs.txt)
Enter your choice: 1
Enter value for a: 1
Enter value for b: -5
Enter value for c: 6

Solving for equation: 1.00x^2 + -5.00x + 6.00 = 0
Output: Two distinct real roots: x1 = 3.00, x2 = 2.00
```

Listing 1: A successful run with direct input.

4.1.2 Example 2: Invalid Input

The program will detect and report all non-numeric inputs.

```
$ ./solver_app
Select Input Method:
...
Enter your choice: 1
Enter value for a: $
Enter value for b: 5
Enter value for c: z
Invalid input: The value of coefficient a = '$' should be numeric
Invalid input: The value of coefficient c = 'z' should be numeric

Calculation aborted due to invalid input.
```

Listing 2: A run demonstrating error reporting.

4.2 Mode 2: File Input

This mode is useful for running predefined test cases without retyping them.

Step 1: Create `coeffs.txt`

In the same folder as your program, create a file named `coeffs.txt`. Inside this file, write the three coefficients, separated by spaces.

Example `coeffs.txt` content:

```
2 4 2
```

Step 2: Run the Executable and Choose

Run the program with `./solver_app`. When prompted, type 2 and press Enter. The program will automatically read the file and show the result.

4.2.1 Example Run with File Input

```
$ ./solver_app
Select Input Method:
...
Enter your choice: 2

Solving for equation: 2.00x^2 + 4.00x + 2.00 = 0
Output: Two equal real roots: x1 = x2 = -1.00
```

Listing 3: A run using the file input method.

5 Troubleshooting

Error: Error opening `coeffs.txt`... This means the program cannot find the input file in Mode 2. Ensure `coeffs.txt` is named correctly and is in the same directory as the executable.

Error: undefined reference to `'sqrt'` This compilation error occurs if you forget the `-lm` flag. Re-compile using the full command: `gcc solver.c -o solver_app -lm`.

Error: gcc: command not found This means a C compiler is not installed on your system. You must install a compiler like GCC (e.g., via MinGW for Windows, Xcode Command Line Tools for macOS, or 'build-essential' on Linux).